IIT MADRAS – PRAVARTAK

OMEGA

Solar Panel Detection & Image Segmentation System

Smarter Vision | Cleaner Energy | A Sustainable Tomorrow

Detect
Solar Installations

Segment
Solar Areas

Enable
Clean Energy

Support
Sustainable Future

Harnessing AI for a Greener Planet

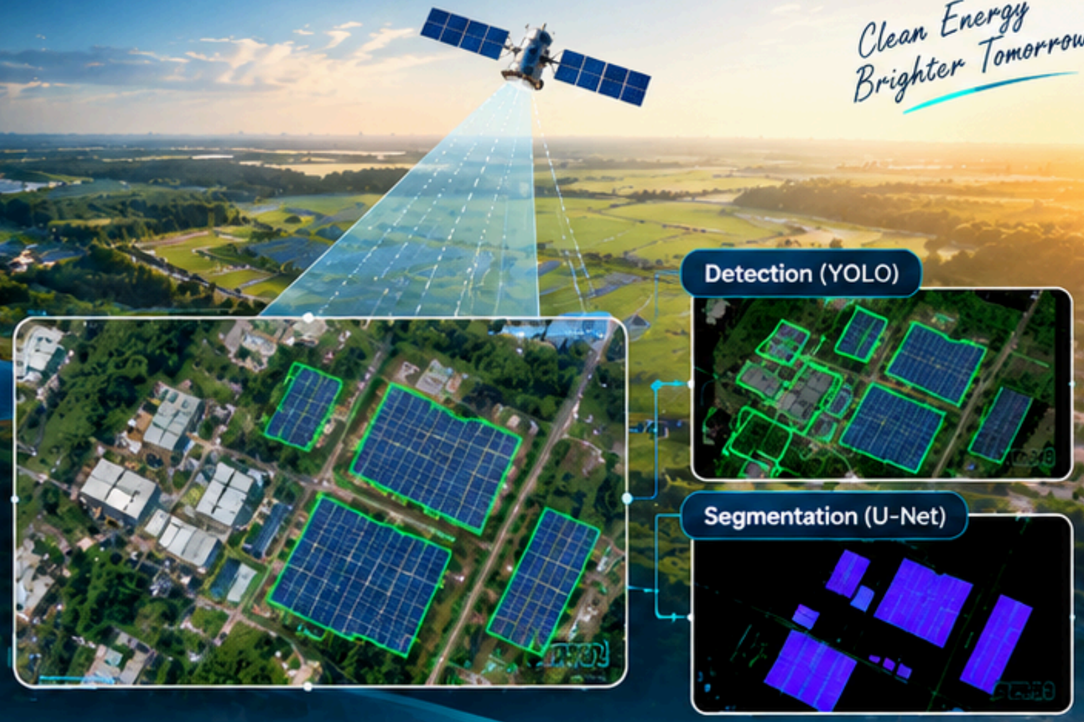
02 PROJECT INTRODUCTION

What is OMEGA?

OMEGA is a computer vision system that **detects** and **segments** solar panels from aerial and satellite images.

It helps in identifying solar installations, supporting renewable energy monitoring, resource planning and sustainable development.

Clean Energy Brighter Tomorrow



Key Objectives

Detect
Find solar installations using bounding boxes (YOLO).

Segment
Highlight solar areas at pixel level (U-Net).

Enable
Support clean energy and efficient resource management.

Build a Sustainable Future
Contribute to a greener and smarter planet.

Why It Matters?

- Solar energy is clean, renewable and essential for a sustainable future.
- Manual inspection is slow, expensive and error-prone.
- AI helps automate detection and provide accurate results at scale.

Aerial & Satellite Images
Wide coverage | High accuracy

AI-Powered Vision
YOLO + U-Net

Real-World Impact
Clean Energy | Sustainable Future

Cleaner Energy Brighter Tomorrow

03 PROBLEM STATEMENT

The Challenge

Identifying and mapping solar installations from aerial and satellite imagery is not easy.

Solar panels are often small, appear in different shapes and orientations, and are mixed with other objects like buildings, roads, trees and fields. This makes manual detection slow, error-prone and not scalable.

From Raw Images to Actionable Insights

Key Problems

- **Small & Scattered Objects**
  Solar panels are small in size and spread across large areas, making them hard to spot in high-resolution images.
- **Similar Surroundings**
  They look similar to other objects like buildings, roads, trees and fields, which leads to false detections.
- **Manual Monitoring is Slow**
  Human inspection is time-consuming, expensive and not scalable for large regions.
- **Need for Automation**
  A reliable and efficient AI system is required to detect and segment solar installations accurately at scale.

OUR GOAL

Build a smart system that can **detect** and **segment** solar panels accurately from aerial and satellite images.

More Renewable Energy | Better Resource Planning | A Greener Future

04 PROPOSED SYSTEM & WORKFLOW

From Image to Insights

A simple yet powerful pipeline to detect and segment solar panels from aerial and satellite images.

01 Input Image

Upload an aerial or satellite image (JPG, PNG, etc.).

Aerial / Satellite Image

02 Preprocessing

Resize, normalize and prepare the image for the models.

Resize • Normalize • Enhance

03 YOLO – Detection

Identify solar panels using bounding boxes.

Bounding Boxes (Solar Panels)

04 U-Net – Segmentation

Generate pixel-level mask for solar panel regions.

Segmentation Mask

05 Results

View detection, segmentation and overlay. Download if needed.

Detection

Mask

Overlay

Original

Key Features

- **Detect** Solar Installations
- **Segment** Solar Areas
- **Enable** Clean Energy
- **Support** Sustainable Future

Models & Tools

- YOLO (Detection)
- U-Net (Segmentation)
- Python
- PyTorch
- Streamlit

Better Solar Mapping for a Greener Planet

OMEGA | Solar Panel Detection & Image Segmentation System

05 Dataset & Preprocessing

Better Data Better Detection

We use the PV03 dataset and apply a series of preprocessing steps to prepare the images and masks for effective model training and evaluation.

1 Dataset

PV03 Dataset
Aerial and satellite images of solar panel installations.

Example Image

Example Mask

Total Images : 2,308
Image–Mask Pairs : 2,308

2 Preprocessing Steps

2.1 Resize

- Resize all images to 640 × 640 (or model input size).
- Maintain aspect ratio (padding if needed).

Original (1024 × 1024) → Resized (640 × 640)

2.2 Normalize

- Scale pixel values to [0, 1].
- Helps with faster and stable training.

Before (0 – 255) → After (0 – 1)

2.3 Contrast & Brightness

- Adjust brightness and contrast for consistent illumination.
- Improves model robustness across different conditions.

Before → After

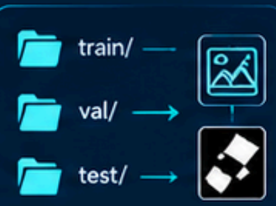
2.4 Data Augmentation

- Horizontal/vertical flip
- Rotation (e.g., 90°, 180°)
- Random crop and zoom
- Color jitter (brightness, saturation)

Original · Flip · Rotate · Zoom

2.4 Prepare for Training

- Split into train / val / test.
- Organize images and masks.
- Verify image-mask alignment.



Final Dataset for Training

Image Size	Dataset Split	Image Format	Mask Format
640 × 640 (after resizing)	70% Train | 15% Val | 15% Test (approx.)	JPG / PNG (RGB)	PNG (Binary / 0-1)

Clean Data → Better Model Performance

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From Image to Bounding Boxes

Target. Localize. Prepare.

Extracting candidate bounding boxes from segmentation masks using OpenCV.

3-STEP WORKFLOW

01 | Input Image

A raw aerial/satellite view of the rooftop installations.

02 | Binary Segmentation Mask

Highlighting the foreground solar regions.

03 | Bounding Box Generation

Contours and bounding boxes using OpenCV.

DETECTION IN ACTION

HOW OPENCV BOUNDING BOXES WORK

Binary Mask Input → Connected-Component Analysis → Contour Extraction → Bounding Box Generation

BENEFITS & KEY FEATURES

No Neural Network
Leverages traditional OpenCV computer vision.

Binary Mask Analysis
Processes connected foreground regions.

Candidate Bounding Boxes
Converts contour detection into precise bounding coordinates.

07 U-Net Segmentation Model

From Pixels to Solar Regions

U-Net is a powerful CNN architecture used for image segmentation.
It helps to identify solar panel regions at the pixel level, giving a clear mask of solar installations.

01 Input Image

- RGB satellite/aerial image
- Resized to 128 × 128 (for U-Net input)

Input Image (128 × 128)

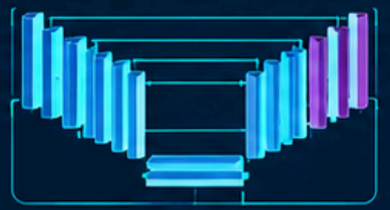
02 Preprocessing

- Resize to 128 × 128
- Normalize (0 – 1)
- Enhance contrast
- (optional: augmentation)

Original → Preprocessed

03 U-Net Architecture

- Encoder (downsampling)
- Bottleneck
- Decoder (upsampling)
- Skip connections



U-Net
(Contracting → Expanding)

04 Training

- Loss Function: Binary Cross Entropy
- Optimizer: Adam
- Epochs: ~50 (example)
- Batch Size: ~8 (example)

Output: Predicted Segmentation Mask (1 × 128 × 128)

05 Output

- Binary mask (0 / 1)
- Highlights solar panel regions
- Remove small noisy regions (optional: morphological ops)

Predicted Mask

06 Overlay Result

- Overlay mask on original image
- Visualize detected solar regions
- Helps in easy interpretation

Overlay (Original + Mask)

U-Net Key Features

U-Net Architecture	Skip Connections	Pixel-level Segmentation	Input Size	Binary Output
(Encoder-Decoder)	(preserves details)	(precise masks)	(after preprocessing)	(0 / 1 mask)

Model Performance (Example)

Loss (BCE)	0.072
Accuracy	98.4%
IoU (Validation)	0.86
Dice Score	0.92

Original Image

Predicted Mask

Identify Solar Regions with Precision

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08 IoU & Dice Score | Model Evaluation

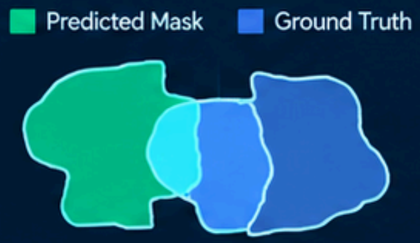
We use IoU (Intersection over Union) and **Dice Score** to evaluate the segmentation performance of our U-Net model. These metrics measure how well the predicted mask matches the ground truth mask.

1 What is IoU?

IoU measures the overlap between the predicted mask and the ground truth mask.

$$IoU = \frac{TP}{TP + FP + FN}$$

TP = True Positive (correctly predicted solar pixels)
FP = False Positive (predicted solar, but not in ground truth)
FN = False Negative (missed solar pixels)



$$IoU = \frac{\text{Overlap Area}}{\text{Union Area}}$$

2 What is Dice Score?

Dice Score is another overlap metric, often used in image segmentation. It gives more weight to the intersection.

$$Dice = \frac{2TP}{2TP + FP + FN}$$

Higher values (closer to 1) indicate better segmentation performance.

Dice Score is more sensitive to small objects and boundary details.

3 Evaluation Process

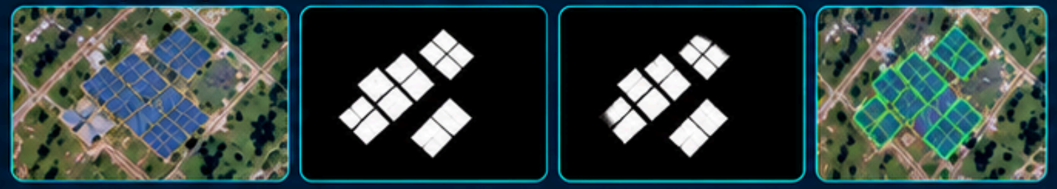
1. Predict mask using U-Net
2. Compare with ground truth mask
3. Calculate IoU & Dice
4. Analyze results & visualize

Evaluation Results

IoU	Dice Score
0.8109 (Test Set)	**0.8680** (Test Set)

High overlap between predicted and ground truth masks

Sample Predictions



Performance Metrics (Test Set)

IoU	Dice Score	Precision	Recall
0.8109	0.8680	0.8923	0.8456

Key Takeaways

- IoU & Dice measure mask overlap and segmentation quality.
- Higher values indicate better performance.
- Our model achieves strong results on the test set.
- These metrics help us fine-tune the model and improve detection accuracy.

Better Data → Better Models → Cleaner Planet

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FINAL SYSTEM RESULTS

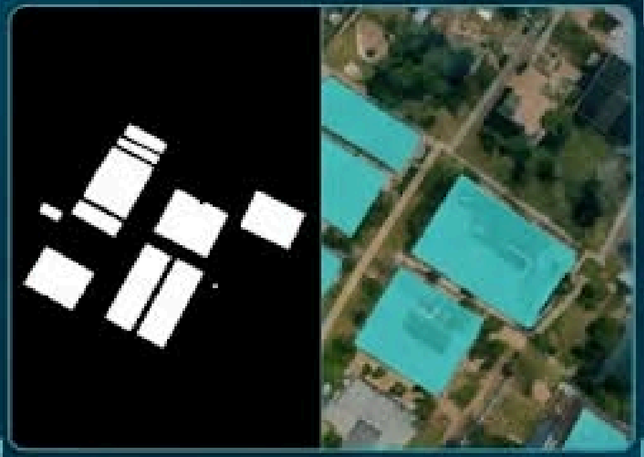
Target. Localize. Evaluate.

Combined OpenCV candidate bounding boxes, U-Net segmentation, and evaluation workflow.

INPUT & BOUNDING BOXES (OpenCV)

OpenCV-Based Candidate Bounding Boxes

U-NET SEGMENTATION



Predicted Segmentation Regions

FINAL MERGED OUTPUT

Visual Analysis Overlay

MODEL EVALUATION METRICS

Intersection over Union
IoU: – (Illustrative)

Dice Similarity Coefficient
Dice Score: – (Illustrative)

BENEFITS & KEY FEATURES

Automated Detection & Segmentation
Unified workflow combining OpenCV bounding boxes and U-Net segmentation.

Visual Localization
Simultaneous validation using boundary and pixel-level localization.

From Satellite Imagery to Solar Panel Localization and Segmentation Evaluation